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Unsupervised Defect Clustering in Atomic-Resolution Microscopy Using a Convolutional Variational Autoencoder

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Abstract

Atomic-scale defects govern many functional properties of materials, yet their systematic identification and quantification remain challenging because supervised learning approaches require extensive labeled datasets, which are scarce in atomic-resolution microscopy due to the complexity and diversity of defect structures. To overcome this limitation, we introduce a fully unsupervised machine learning framework capable of discovering and clustering defect structures without prior labeling or predefined defect classes. The framework employs a convolutional variational autoencoder (CVAE) to reconstruct ideal, defect-free images, enabling the generation of difference images that isolate local structural anomalies. From these, 47 features are extracted and refined through a three-tier feature selection process to minimize redundancy and noise. Dimensionality reduction via principal component analysis (PCA), combined with silhouette score optimization, guides the determination of the optimal cluster number prior to applying k-means clustering, which yields well-separated groups corresponding to distinct defect types. Validated on CdTe and $SrTiO_3$ datasets, this unsupervised, label-free approach enables high-throughput defect discovery and clustering in scanning transmission electron microscopy (STEM) and related imaging modalities.

Introduction

Advancements in materials science rely heavily on the precise characterization of structural defects, as these imperfections critically influence material properties[1–3]. Defects, such as vacancies, interstitials, substitutions, antisites, adatoms, or grain boundaries have been extensively studied in both 2D and 3D materials due to their profound impact on electronic, optical, and optoelectronic properties[1,2,4–10]. Depending on their characteristics, defects can act as carrier donors[10], traps[1], scattering centers[9], or recombination sites[2], directly affecting material performance. Defect engineering enables the systematic modification of material properties by eliminating detrimental defects or introducing beneficial ones, thereby optimizing the performance of electronic and optoelectronic devices[11–14]. For atomic-scale characterization of such defects, aberration-corrected scanning transmission electron microscopy (STEM) has emerged as an essential technique, providing direct atomic-resolution imaging of material structures[3,15–19]. While STEM allows unparalleled visualization of structural anomalies, the increasing volume of data generated renders manual analysis highly labor-intensive, subjective, and prone to human bias.

Consequently, integrating artificial intelligence (AI) and machine learning (ML) into microscopy workflows has become imperative for achieving automated, high-throughput data analysis[20–25].

AI-driven approaches leverage the periodic nature of crystalline materials to distinguish anomalous regions from bulk defect-free areas, significantly enhancing defect identification. However, conventional ML models primarily rely on supervised learning, which presents two major limitations: (i) the need for large-scale pre-labeled datasets, which are often impractical to generate for novel materials or rare defect structures, and (ii) high computational costs, necessitating access to high-performance supercomputing resources for training complex models[23,26]. These challenges have spurred growing interest in unsupervised machine learning techniques, which can autonomously extract meaningful features and cluster images without requiring manual annotations, thereby streamlining microscopy data management and analysis[27,28].

Among unsupervised machine learning models, variational autoencoders (VAEs) have emerged as powerful tools for extracting low-dimensional, informative representations from high-dimensional data[29,30]. Specifically, convolutional variational autoencoders (CVAEs) have demonstrated exceptional capabilities in reconstructing defect-free images after being trained on a single, reasonably sized STEM image containing at least ~20 × 20 unit cells[31–33]. Their use in materials science continues to grow, with recent studies refining training strategies to improve latent representations and learning performance across complex experimental datasets. These capabilities have inspired us to explore the potential of clustering image data based on defect types, enabling automated defect identification and efficient microscopy data management. If a set of well-defined features can be extracted from input images by leveraging the predictive capabilities of CVAEs, defect identification can be fully automated, removing human intervention from the clustering process.

Here, we propose an unsupervised machine learning framework that autonomously identifies structural anomalies in atomic-resolution images using a convolutional variational autoencoder (CVAE), followed by the extraction of 47 general features and their refinement through a three-tier feature shortlisting process. Once the refined features are obtained, the optimal number of clusters (N_c) and principal component (PCA) dimensions are determined by maximizing the silhouette score (S) to classify defects into distinct clusters, followed by t-SNE visualization for visual representation of the clustering results. Our approach integrates deep learning with feature-based clustering to uncover intrinsic patterns in defect structures, providing a robust pathway for clustering, managing, and analyzing large image datasets based on the types of structural anomalies they contain. The integration of this unsupervised learning model into microscopy not only enhances defect detection accuracy but also paves the way for fully automated, AI-driven materials characterization, accelerating the discovery and optimization of functional materials.

Importantly, the framework is generalizable and can be applied to any material with a periodic structure. To validate its efficacy, we applied the model to two well-known materials, cadmium telluride (CdTe) and strontium titanate ($SrTiO_3$), demonstrating its robustness and accuracy in defect identification and clustering. The proposed model is computationally efficient and can be executed on standard laboratory workstations without requiring specialized hardware or machine learning libraries beyond commonly available Python distributions (see Supplementary Section A for details). This accessibility makes the framework well-suited for routine laboratory use, enabling real-time, automated defect identification and clustering in atomic-resolution microscopy sessions.

Results

Unsupervised Clustering Framework Enabled by CVAE-Based Anomaly Detection

A vast amount of image data containing diverse structural anomalies is generated daily during microscopy sessions. Labeling such large datasets for training supervised machine learning models is impractical and insufficient, as new and previously unseen types of defects can appear at any time. To overcome these limitations, we employ an unsupervised clustering approach to find structural variations in atomic-resolution images and group them based on the distinct types of defects they contain.

The overall clustering workflow of the proposed unsupervised machine learning framework is illustrated in Fig. 1 (a). The process begins by training a CVAE on a bulk, defect-free region of the target material to learn its underlying periodic lattice structure. As shown in Fig. 1 (b), the CVAE follows an encoder–decoder architecture to capture the latent space representation of the material structure, where the encoder and decoder are mirror networks composed of three convolutional and three deconvolutional layers, respectively. During training, the encoder maps each input image x to the parameters $\mu(x)$ and $\log\sigma^2(x)$, where μ represents the latent mean vector and σ^2 represents the corresponding latent variance, defining a Gaussian latent distribution. Latent vectors (z) are sampled using the reparameterization

$$z = \mu + exp\left(\frac{1}{2}log\sigma^2\right) \odot \epsilon \tag{1}$$

where $\epsilon \sim \mathcal{N}(0, I)$ is a random noise vector drawn from a zero-mean, unit-variance normal distribution, and I denotes the identity covariance matrix. This formulation enables backpropagation through the stochastic sampling process. The decoder reconstructs a defect-free image from z, while training minimizes the evidence lower bound (ELBO), consisting of a reconstruction loss and a Kullback–Leibler (KL) divergence term that regularizes the latent distribution toward a standard normal prior, ensuring both accurate image generation and smooth, continuous latent-space learning (see the Methods section for a detailed discussion of the CVAE architecture, and Supplementary Sections A and B for details on the CVAE inputs, outputs, training and sensitivity analysis).

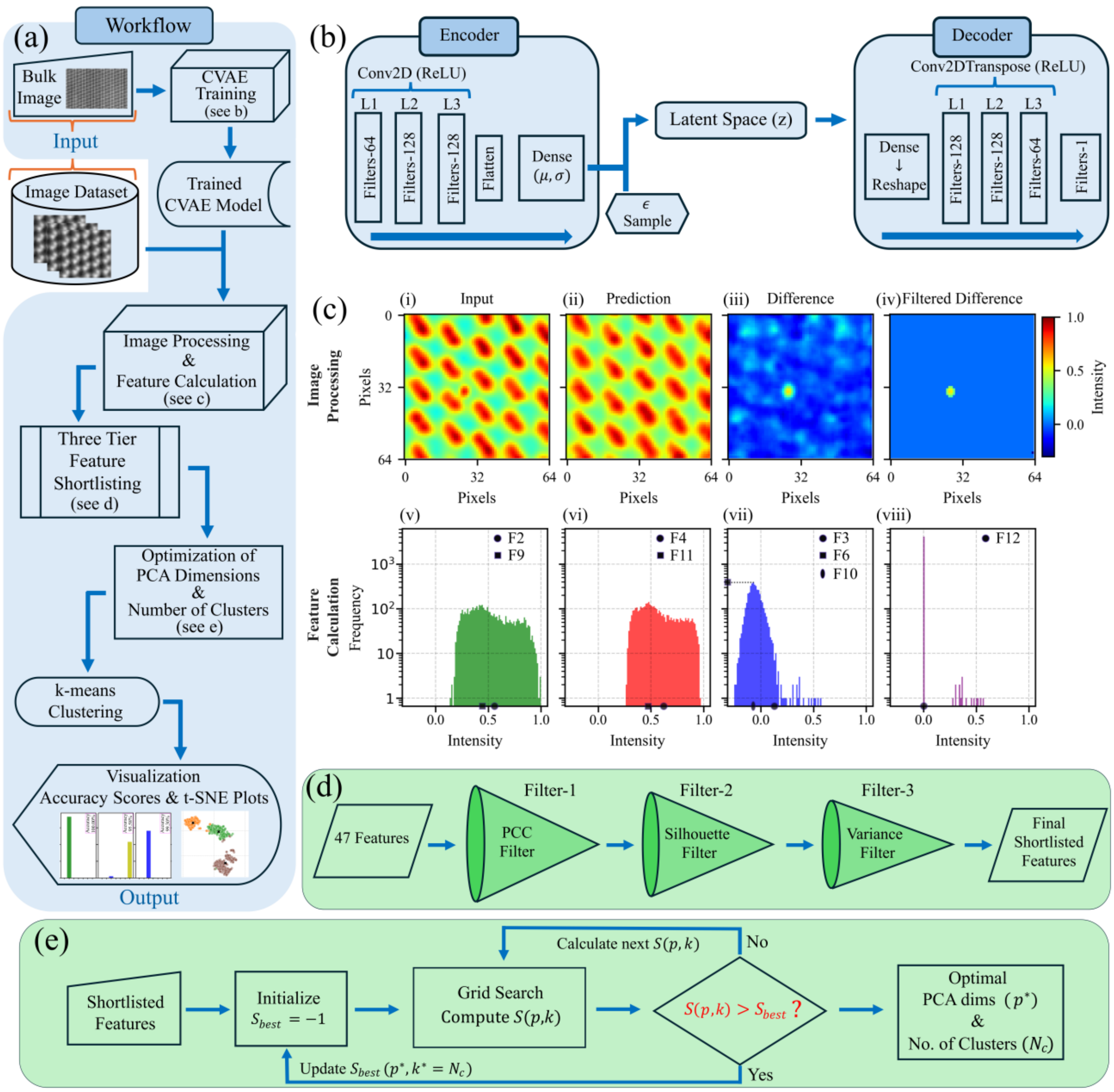

Figure 1: Overview of the clustering pipeline. (a) Flowchart illustrating the complete pipeline, from CVAE training on periodic lattice structures to clustering based on identified anomalies. **(b)** CVAE architecture comprising an encoder-decoder pair used to learn latent representations of the periodic structures. **(c)** Illustration of how the trained CVAE iteratively processes all images in the dataset, exemplified by an image containing an interstitial atom: (i) Colored grayscale input, (ii) defect-free reconstruction by the CVAE, (iii) difference image, and (iv) noise-filtered difference obtained by thresholding. (v–viii) Corresponding histograms of pixel-intensity distributions for the input, predicted, difference, and filtered difference images, used to extract 47 quantitative features. **(d)** Three-tier feature shortlisting process, removing noisy and uninformative features before clustering. **(e)** Workflow for determining the optimal number of clusters (N_c) and PCA dimensions (p^*) by maximizing the silhouette score (S).

For this study, the CVAE was trained on a portion of the atomic-resolution image presented in Fig. 2(a). Once trained, the CVAE reconstructs a defect-free image for each input image, enabling reconstruction-based anomaly detection by identifying deviations associated with structural defects, thereby transforming defect identification into a difference-based anomaly detection problem, as illustrated in Fig. 1(c). Hence, the CVAE approach (i) learns the normal bulk distribution and flags statistically unlikely deviations as defects, (ii) enables discovery without prior knowledge of defect type, and (iii) reframes the task from learning all possible defects to learning what perfect looks like. From the intensity distributions of the input, predicted (reconstructed), difference, and filtered difference images shown in the second row of Fig. 1(c), 47 quantitative features are extracted, capturing geometric, intensity, frequency, and periodicity-based characteristics of the atomic-resolution images (see Supplementary Section B.iii for a detailed description of the 47 features). Selected features (F2–F4, F6, and F9–F12) are highlighted in Fig. 1(c)(v–viii). As depicted in Fig. 1(d), these features undergo a three-tier filtration process to remove redundant and noise-inducing parameters, followed by principal component analysis (PCA) for dimensionality reduction. The reduced feature space is then clustered using the k-means algorithm, with the optimal number of clusters and PCA dimensions determined by maximizing the silhouette score (S), as demonstrated in Fig. 1(e), using a grid search across a predefined range of cluster counts and PCA dimensions (up to 20). Finally, t-distributed stochastic neighbor embedding (t-SNE) provides a two-dimensional visualization of the clustering results, offering a qualitative illustration of the separation between distinct defect types within the dataset. The computational cost and system requirements for the complete workflow, from CVAE training through clustering, are provided in Supplementary Section A.ii, confirming that the workflow is computationally efficient and practical for standard laboratory workstations. A detailed discussion of each step is provided in the following sections.

The image shown in Fig. 2(a) was specifically selected to demonstrate the performance of our clustering model, as it contains seven visually distinct regions. Five of these correspond to real, experimentally observed image types, including four defect types and one defect-free bulk region (Bulk), while the remaining two represent artificially introduced defects. Figs. 2(b) and 2(c) illustrate artificially generated defects simulating interstitial (Int) atoms and vacancies (Vac), respectively. Figs. 2(d)–2(h) correspond to type-1 stacking fault (SF-1), a Lomer–Cottrell (L–C) dislocation configuration which appears at the intersection of two distinct stacking faults[34], defect-free bulk region, type-2 stacking fault (SF-2), and a coherent twin boundary (TB). While no detailed crystallographic analysis is pursued here, the differences in contrast, orientation, and morphology allow us to distinguish SF-1 and SF-2 as visually and structurally distinct features, which are referenced accordingly throughout the text. This categorization provides a diverse and representative set of structural motifs for evaluating the clustering performance of our framework.

A major challenge in clustering such images using ML and computer vision techniques arises from variations in edge effects, which occur due to image translation (window shifts). When an image window is translated, identical image types, such as a bulk region, may begin at different positions within the atomic lattice, for instance, starting at the center or edge of an atomic column. As a result, identical image types often exhibit different edge effects, i.e., bulk and interstitial (Int.)

images, as demonstrated in Figs. 2(i), (j) and Figs. 2(k), (l). Conversely, images containing different types of defects may share similar edge effects, as illustrated by the pairs in Figs. 2(i), (k) and Figs. 2(j), (l), which can lead to incorrect clustering by conventional computer vision models. Such models may inadvertently prioritize edge shapes over actual defect features, misclassifying images that contain half or full atomic columns (arising from translation) as distinct image types. Therefore, it is crucial to develop a model capable of disregarding edge effects while effectively clustering images based on their intrinsic defect structures. To address this, we propose a refined

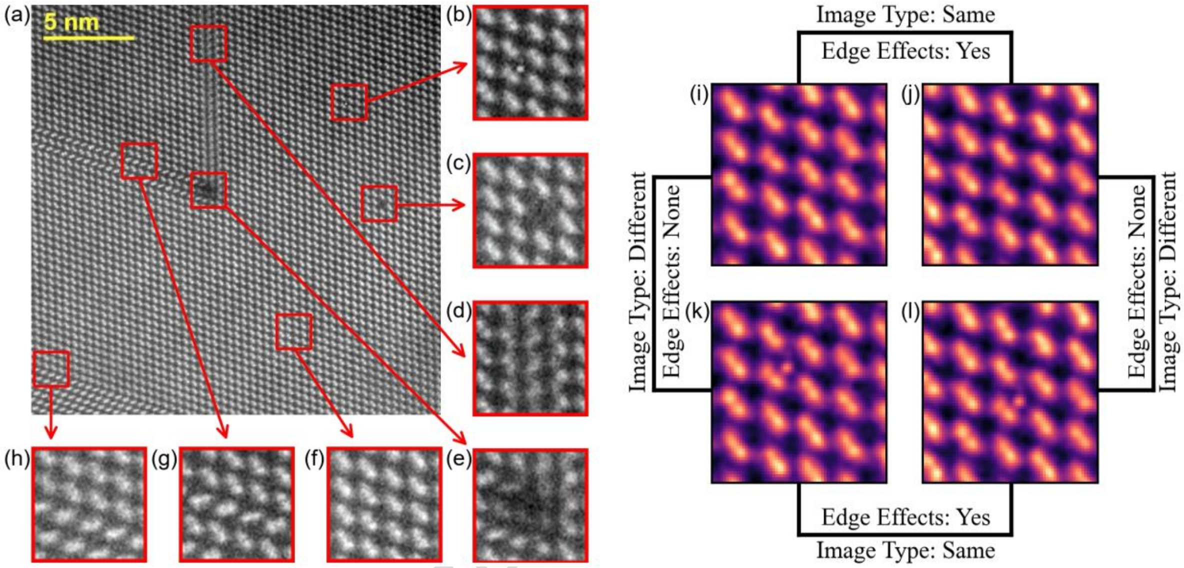

Figure 2: Representative Atomic-Resolution STEM Image of CdTe with Annotated Defects and Edge Effects. (a) Atomic-resolution high-angle annular dark field (HAADF) image of CdTe along the [110] crystallographic direction. The scale bar represents 5 nm. The HAADF STEM image was acquired using an aberration-corrected cold field emission scanning transmission electron microscope operated at an accelerating voltage of 200 kV. The field of view spans approximately 22 × 22 nm, where the CdTe dumbbells are distinctly discernible, exhibiting a pronounced bright contrast. The inner and outer collection angles for the HAADF detector were set to 68 mrad and 280 mrad, respectively. The image has dimensions of 768 × 768 pixels, with seven distinct regions highlighted by red boxes of size 64 × 64 pixels. The zoomed-in views of these regions are presented in **(b)** an interstitial atom, **(c)** a missing CdTe dumbbell, **(d)** a type-1 stacking fault, **(e)** a Lomer-Cottrell dislocation, **(f)** a defect-free bulk region, **(g)** a type-2 stacking fault, and **(h)** a twin boundary. **(i)–(l)** illustrate the presence of translation-induced edge effects in the images used for clustering analysis: **(i)** and **(j)** correspond to the same image type (defect-free bulk region) but with different edge effects, while **(k)** and **(l)** represent the same image type (interstitial atom) but with varying edge effects.

feature extraction approach based on CVAE that prioritizes defect characterization over edge artifacts, thereby improving the robustness and accuracy of defect identification and clustering. This clustering was enabled by CVAE architecture, which has proven effective for anomaly detection in high-dimensional datasets[31,32,35]. To quantify the influence of these edge effects, we performed an ablation study by clustering directly on raw input images without using the CVAE (see Supplementary Section E for details).

Feature Extraction

To perform effective clustering, it is crucial to define meaningful features that capture key insights and intricate details of the data. Hence, we formulated these features based on the pixel intensity distributions of the input, predicted, difference, and filtered difference images. Once the CVAE model is presented with an input image, it generates a corresponding predicted image. If the model has been effectively trained, the predicted image should closely resemble a perfect bulk image, indicating that the model has successfully learned the underlying periodic structure of the material. By subtracting the predicted image from the input image on a pixel-by-pixel basis, a post-processing step constructs the difference image, thereby highlighting deviations from the pristine structure. To further refine the results and eliminate noise, a filtered difference image is generated by applying a thresholding technique, in which pixel intensities below a predefined value are discarded (see Supplementary Sec. B-i for details). This processing step enhances the visualization of detected anomalies by minimizing irrelevant intensity fluctuations. Consequently, the filtered difference image contains only the defect-related information, effectively excluding undesired contributions such as edge effects and other non-structural artifacts. Thus, the use of filtered difference images enables the model to focus solely on intrinsic anomalies while eliminating peripheral artifacts. As shown in Fig. 1(c), the model successfully identifies the interstitial atom within the input image. A comparative analysis of the input, predicted, difference, and filtered difference images in Figs. 1(c)(i-iv) clearly demonstrates the model's step-by-step capability in detecting anomalies in (S)TEM images. Notably the filtered difference image does not contain structural information about the original image but rather contains only the anomaly-related information; hence, the variations in edges (edge effects) introduced by image translation are effectively suppressed. This suppression, in turn, allows the model to cluster images solely based on intrinsic anomalies rather than peripheral edge features. Furthermore, Figs. 1(c)(v-viii) illustrate the variations in pixel intensity distributions and corresponding frequencies across these four image types. This trend is further validated in Fig. S5, where similar variations in pixel intensities are observed across the remaining six image categories. Notably, these variations are significant not only among the input, predicted, difference, and filtered difference images (intra-image differences) but also across the seven distinct image categories (inter-image differences), as highlighted in the last column of Fig. S5.

These observed variations in pixel intensities and their distributions across different image types exhibiting distinct anomalies, provide a robust foundation for defining generalized clustering features applicable to all crystalline periodic structures. By leveraging these features, images can be clustered based solely on the defects they contain, eliminating the need for prior knowledge or labelling of specific defect types. This approach enables the implementation of unsupervised machine learning, allowing for automated and unbiased clustering of defect structures.

Initially, a total of 47 features were defined. These extracted 47 features encapsulate various characteristics of input, predicted, difference, and filtered difference images, offering a comprehensive analysis of the images. Intensity-based features include midpoints of intensity distributions, frequency and amplitude of the most recurrent intensities, as well as statistical metrics such as skewness, kurtosis, and standard deviation, which collectively provide insights into brightness distribution and contrast variations. Edge-based features, including edge count and length, capture structural details and discontinuities. Radial symmetry evaluates spatial consistency in patterns, while frequency-domain features, such as the mean and variance of Fast Fourier Transform (FFT) coefficients, analyze texture and periodicity. Integrated intensity measures track total positive and negative intensities, facilitating the identification of bright and dark regions. Additionally, shape-based features, such as the largest connected region, highlight dominant structures, while Fourier band ratios assess intensity distribution across different frequency bands, enriching the feature space for image analysis and comparison. For a comprehensive list of features and their definitions, refer to Supplementary Section B-iii.

Following the extraction of all 47 features from the CVAE model output, the feature dataset was normalized to a range of −1 to 1. This normalization ensures uniform scaling across all features, mitigating biases and enhancing the reliability of subsequent analyses. In clustering-based machine learning, assessing feature distributions and variations is essential to determine their effectiveness in distinguishing meaningful clusters. To quantitatively evaluate the variation in feature distributions, box plot analysis was conducted for all 47 features. The feature values were analyzed across seven distinct image types in the CdTe dataset, which comprises 1120 atomic-resolution STEM images. Each box plot represents the distribution of a specific feature across these image types, as illustrated in Supplementary Section C (Figs. S6 and S7). The box plot visualizations reveal significant variance in feature values across different image types, a critical indicator of their discriminative capability. This variation confirms that the selected features encode meaningful structural differences, reinforcing their suitability for unsupervised clustering.

Three-Tier Feature Selection and Dimensionality Reduction for Clustering

Building on the feature distribution analysis, a systematic three-tier feature selection process was implemented to refine the dataset and enhance clustering performance, as outlined in the feature selection workflow in Fig. 1(d). In feature-based clustering, distinguishing informative features from redundant or noisy ones is critical for optimizing algorithmic efficiency and ensuring robust clustering performance. To achieve this, Pearson correlation coefficient (PCC) analysis was first applied to identify highly correlated features as defined in Eq. (2)[36,37].

$$r = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2 \sum_{i=1}^{n}(y_i - \bar{y})^2}} \tag{2}$$

$$S = \frac{1}{n}\sum_{i=1}^{n}\frac{b(i) - a(i)}{\max\{a(i), b(i)\}} \tag{3}$$

$$\sigma^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2 \tag{4}$$

where x_i and y_i are the individual values of variables X and Y in the sample, $\bar{x}$ and $\bar{y}$ are their respective means, and n is the total number of samples. The Pearson correlation coefficient r measures the linear relationship between X and Y. In the silhouette formulation (Eq. (3)), $a(i)$ denotes the average intra-cluster distance for sample i, and $b(i)$ is the smallest average inter-cluster distance between that sample and points in neighboring clusters, while in Eq. (4), σ^2 represents the variance of variable X.

Figs. 3(a) and 3(b) show box-and-whisker plots for features F1 and F5, which exhibit nearly identical distributions, indicating that both features convey similar information. To further validate this observation, their linear dependency is plotted in Fig. 3(c), where overlapping curves correspond to r = 1. Similarly, Figs. 3(e) and 3(f) present box-and-whisker plots for features F25 and F39, which also display closely matching patterns, as confirmed by their near-perfect linear dependency in

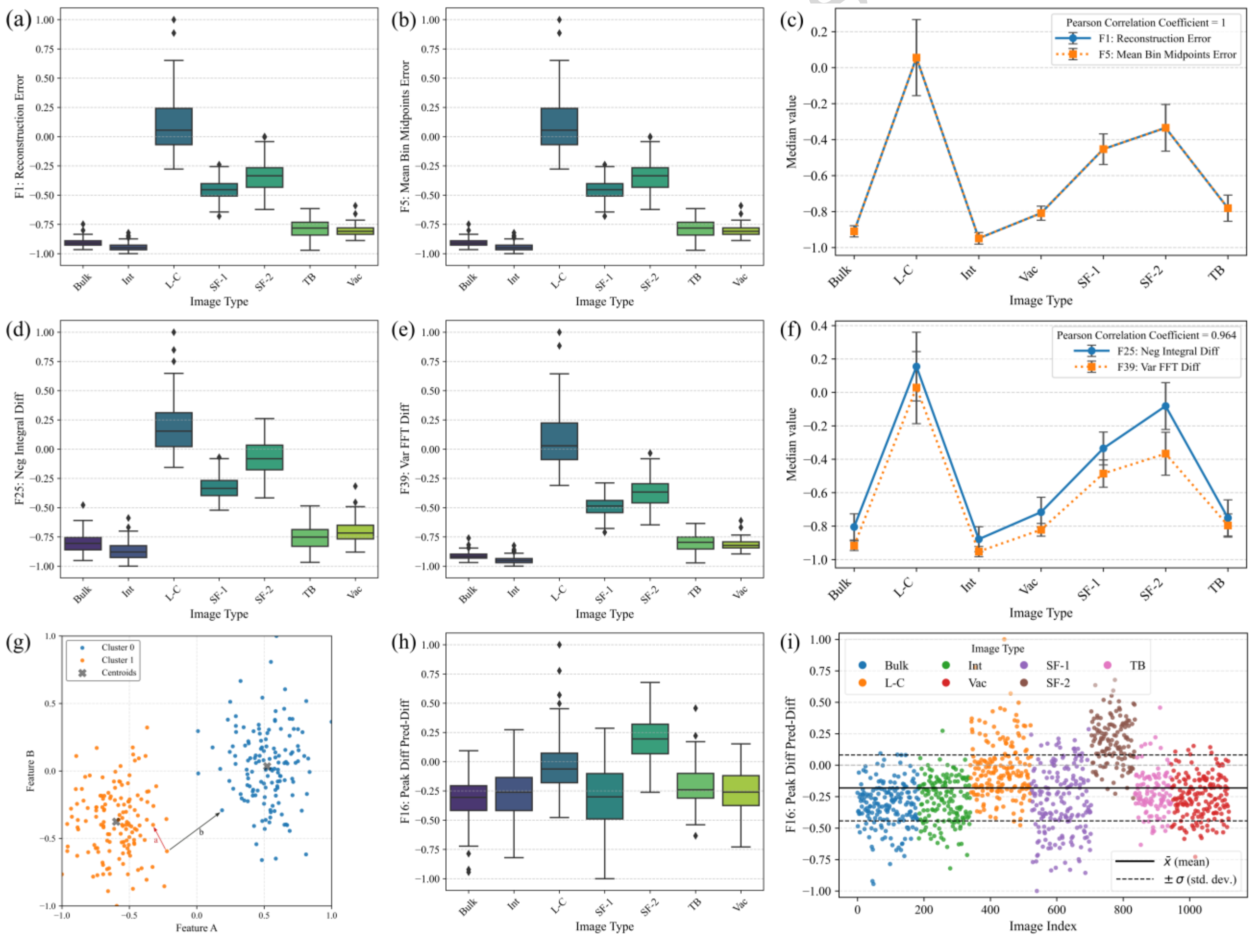

Figure 3: Feature filtering analysis. (a,b) Box-and-whisker plots of features F1 and F5 showing identical distributions, indicating redundant information content. **(c)** Linear-dependency plot for F1 and F5 with correlation coefficient r = 1. **(d,e)** Box-and-whisker plots of features F25 and F39 displaying similar trends, confirmed by their near-perfect linear correlation in **(f)** (r = 0.96). **(g)** Illustration of the silhouette-score geometry, where a represents the average intra-cluster distance and b the smallest average inter-cluster distance used in Eq. (3). **(h,i)** Box-and-whisker

and value-distribution plots of feature F16, illustrating a narrow variance range (± σ = 0.32) across all image types, indicating negligible contribution to clustering.

Fig. 3(f) (r = 0.96). Features with such high correlation coefficients are therefore regarded as redundant, since they provide overlapping information about the dataset. The inclusion of highly correlated features has been shown to reduce model performance and increase computational cost; hence, an effective feature-selection procedure should ideally retain only one representative feature from each correlated group[38,39]. The PCC matrix for all 47 normalized features prior to shortlisting is presented in Fig. S8. A correlation threshold of ± 0.95 was set, ensuring that when two features exhibited a PCC greater than $|0.95|$, one was eliminated to mitigate redundancy. Next, silhouette score (S), as defined in Eq. (3) and illustrated in Fig. 3(g), thresholding was employed to discard features with $S \leq 0$ or those incapable of contributing at least two distinct clusters. When the average intra-cluster distance *a(i)* exceeds the smallest average inter-cluster distance *b(i)*, the data points become intermixed without clear cluster boundaries, indicating that the corresponding feature fails to capture meaningful structural variations in the dataset[40,41]. The results of this filtering step are shown in Fig. S9. Finally, a variance threshold filter, as defined in Eq. (4), was applied[42], retaining only features with a variance score (σ^2) greater than 0.1, since low-variance features typically introduce noise rather than contributing valuable clustering insights. For example, as illustrated in Figs. 3(h) and 3(i), the values of feature F16 mostly lie within a narrow range ($\pm\sigma = 0.316$) for all image types, as indicated by the dashed lines in Fig. 3(i). This limited spread demonstrates minimal variability in the feature values, implying that such a feature cannot contribute meaningfully to cluster separation or structural distinction. Therefore, a variance threshold of 0.1 was adopted, consistent with common practice in ML–based feature-selection algorithms. The variance distributions before and after applying this threshold are presented in Figs. S10 (a) and S10(b), respectively. The correlation matrix of the remaining 33 features produced by this approach is displayed in Fig. S11. Further details on the feature selection methodology are provided in Supplementary Section D.

The impact of the three-tier feature selection process is evident from the comparative analysis in Fig. 4(a), which contrasts clustering performance before and after feature shortlisting for both CdTe and $\mathrm{SrTiO_3}$. The results show that this refinement not only reduced the number of features (F) from 47 to 33 for CdTe and 29 for $\mathrm{SrTiO_3}$ but also led to a substantial improvement in clustering accuracy in both datasets. In Fig. 4(a), a direct comparison of feature count (F), optimized number of clusters (N_c), silhouette scores (S), and clustering accuracy (A) before and after filtering highlights the importance of eliminating redundant and non-informative features. This refinement enhances clustering robustness, demonstrating that a systematically curated feature set is crucial for achieving optimal cluster separability and extracting meaningful data-driven insights.

Determination of Optimal Number of Clusters and PCA Dimensions

Once the final shortlisted feature set was established, we aimed to automatically determine the optimal number of clusters (N_c) and the optimal principal component dimensions (p^*) for clustering. PCA was utilized to optimize both the number of clusters and the dimensionality of the transformed feature space, using S as a validation metric. PCA facilitates dimensionality reduction while preserving the most significant variance, thereby improving clustering efficiency[43,44]. To determine p^*, the number of principal components (p) was systematically varied, and clustering performance was evaluated across different subspaces. The k-means algorithm was applied to the PCA-transformed data, with the number of clusters (k) varying within a predefined range of 2 to 12. To preserve most of the dataset's variance while avoiding overfitting or redundant components, only PCA dimensions retaining at least 90% of the cumulative variance were considered for clustering. The score S, which quantifies cluster compactness and separation, was computed for each combination p and k. A grid search was then performed over p =2 – 20 and k = 2 – 12 to identify the optimal combination (p^*, N_c) that yielded the maximum S. This iterative optimization ensured an optimal balance between data representation and clustering effectiveness, as illustrated

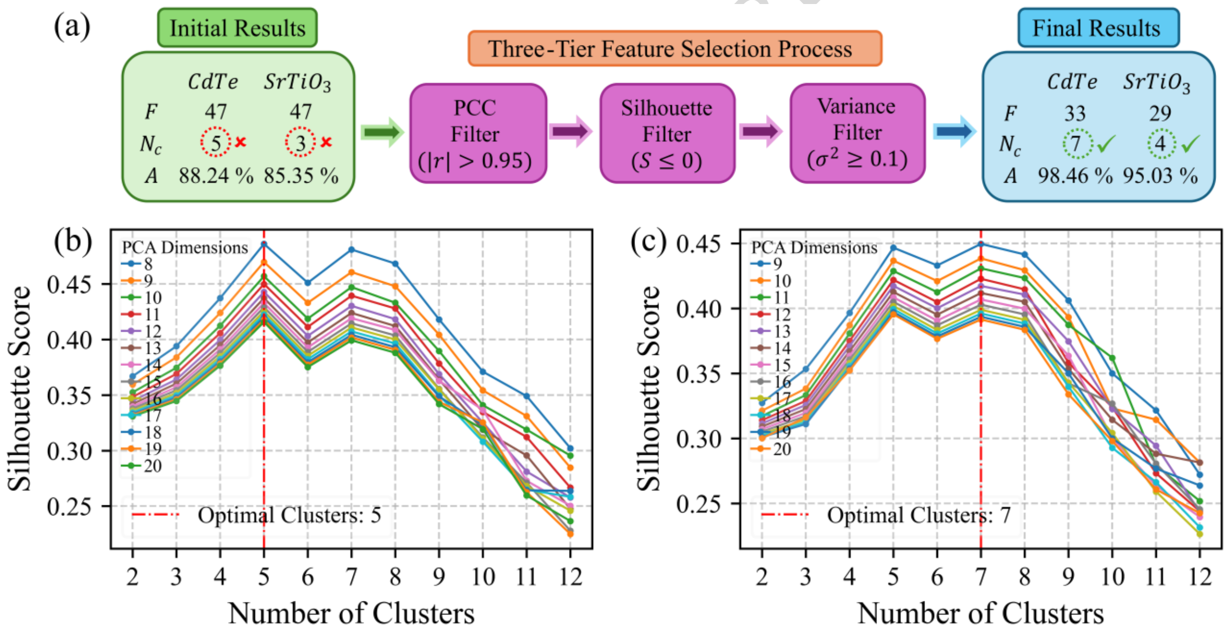

Figure 4: Feature Selection and Automated Clustering Optimization for Atomic-Resolution STEM Images. (a) Schematic of the feature selection pipeline, illustrating descriptor refinement and clustering enhancement for atomic-resolution STEM images. The flowchart tables summarize results before and after three-tier feature selection, where F is the number of features, N_c is the optimal number of clusters, and A represents overall clustering accuracy. **(b), (c)** Automated optimization process for determining the optimal PCA dimensions and number of clusters for the CdTe dataset, before and after the three-tier feature selection process, respectively, utilizing silhouette score maximization as the key criterion. The red vertical dashed lines indicate the optimal number of clusters.

in the workflow shown in Fig. 1(e). Corresponding grid search heatmaps are provided in the Supplementary Section F. By filtering out irrelevant or noise-prone features, PCA enhanced the separability of data points while reducing computational complexity, making clustering more efficient.

The optimization graphs for CdTe, which depicts S as a function of the number of clusters and PCA dimensions, are shown in Figs. 4(b) and 4(c). In Fig. 4(b), where all 47 features were used prior to the application of the three-tier feature selection process, N_c, determined by maximizing S, was found to be five. This value deviates from the actual number of image types present in the dataset (seven). However, after implementing the three-tier feature selection process, as shown in Fig. 4(c), an iterative optimization over both the k and p yielded N_c equals to seven and a corresponding PCA dimensionality of nine, both determined automatically by the maximization of S. These results indicate that the feature selection procedure effectively removed non-informative and noisy features, thereby enabling the unsupervised algorithm to accurately recover the true number of clusters without any prior knowledge. Once the optimal PCA dimensions and number of clusters were established, further analyses were conducted to visualize clustering results and assess accuracy. The PCA transformation was applied to the normalized shortlisted feature set, effectively reducing dimensionality while maintaining cluster quality, and data separability, and preserving maximum variance. Clustering was performed using k-means, assigning each data point to a specific cluster label.

Visualization and Quantitative Assessment of Clustering

To evaluate clustering performance, two primary methods were employed. First, to visualize the clustering of all 1120 images, the t-SNE algorithm was applied to the PCA-transformed data, further reducing it to two dimensions while preserving local data structure. Two t-SNE plots were generated: one illustrating cluster assignments (Fig. 5(a)) and another depicting image types based on predefined labels extracted from image filenames in the initial dataset (Fig. 5(b)). In the cluster-based t-SNE plot (Fig. 5 (a)), each data point (image) was color-coded according to its assigned cluster, providing a visual representation of cluster separation. In the image-type-based t-SNE plot (Fig. 5(b)), different markers and colors were used to distinguish the seven image categories, facilitating a qualitative assessment of whether similar image types were appropriately grouped within the same clusters. It can be seen in Figs. 5(a) and 5(b) that the clusters are well-structured around their respective centroid points, which are indicated by cross (×) markers in the plots. Furthermore, the seven image types have been distinctly categorized into seven separate clusters, confirming that the clustering process has effectively captured the inherent structure of the dataset.

Second, to quantitatively assess clustering performance, an external validation approach was employed by analyzing image type distributions within each cluster. Since the dataset contained labeled images solely for validation purposes, these labels were used to compute the accuracy of individual clusters, as shown in Fig. 5(c). The accuracy of each cluster, denoted as A_c, is defined as the proportion of images belonging to the most frequent image type within that cluster. The overall clustering accuracy, A, is then computed as the mean accuracy across all clusters, expressed mathematically as

$$A_c = \frac{\max_i(n_{i,c})}{N} \times 100 \quad , \quad A = \frac{1}{K}\sum_{c=1}^{K} A_c \tag{5}$$

where c represents a specific cluster, $n_{i,c}$ denotes the number of images of type i in cluster c, and $\max_i(n_{i,c})$ identifies the most frequent image type in cluster c. Additionally, N refers to the total number of images in cluster c, while K represents the total number of clusters. This metric provides a quantitative measure of how effectively the clustering algorithm groups similar image types. A detailed comparison of both the silhouette score and the external accuracy metrics is provided in the Supplementary Section I.

As shown in Fig. 5(c), histograms were generated for each cluster to illustrate the distribution of image types, with accuracy values annotated. These histograms enabled further inspection of misclassified images and potential overlaps between different image types. The final clustering assignments and accuracy scores were used to evaluate the effectiveness of the selected PCA dimensions and cluster count, ensuring that the clustering results were both statistically and visually meaningful. The final overall accuracy achieved for 1120 CdTe images, comprising seven image types and seven automatically determined clusters, is 98.46%, a substantial improvement

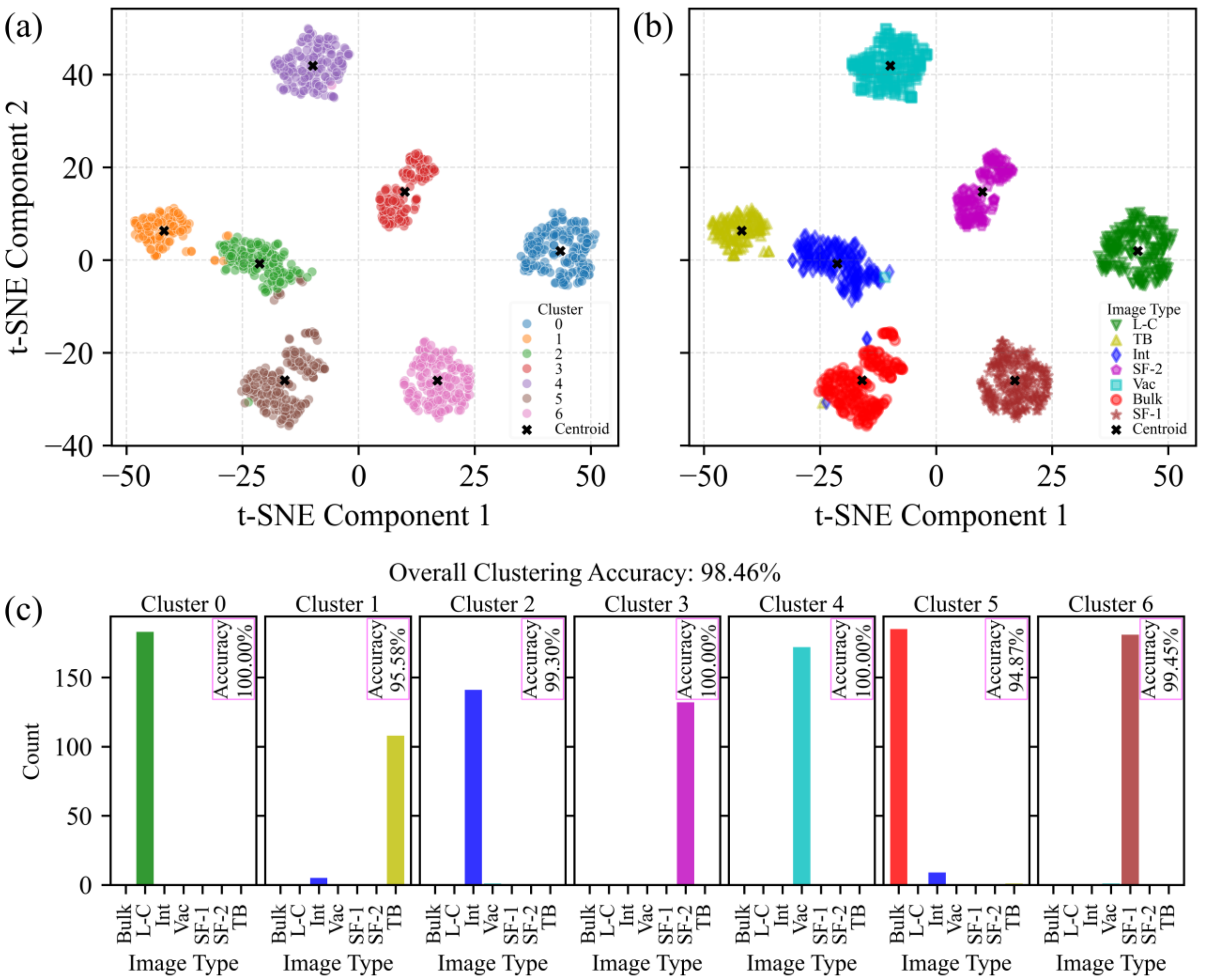

Figure 5: t-SNE visualization of feature distributions for a dataset comprising 1120 atomic-resolution STEM images of CdTe, categorized into seven distinct image types. (a) Clustering-based t-SNE representation, where data points are colored according to their assigned clusters. Cluster centroids are marked with black "x" symbols, illustrating well-separated feature groupings. **(b)** Image-type-based t-SNE representation, where data points are assigned colors and markers corresponding to their image types, confirming that similar images are grouped within the same cluster. Centroids remain consistent across both visualizations, underscoring structural similarities. **(c)** Histogram representation of clustering accuracy, demonstrating an overall accuracy of 98.46%.

over the 88.24% accuracy achieved prior to applying the three-tier feature selection process (Fig. S17). This indicates that only 17 images out of 1120 were misclassified, demonstrating the robustness and effectiveness of the clustering algorithm in identifying defects in S(TEM) images and grouping them based on defect types. The clustering results obtained using all 47 features before applying the three-step feature selection process are presented in Supplementary Section G.

Generalization to $SrTiO_3$: Validation on a Second Dataset

To further assess the efficacy and confirm the generalizability of the developed clustering technique, we applied the same model to the $SrTiO_3$ dataset, which comprises 764 images of 64 × 64 pixels, categorized into four distinct types, as illustrated in Fig. 6 (a). The interstitial (Int), substitutional (Sub), and vacancy (Vac) defects within the images were artificially introduced to enhance dataset variability. The corresponding predicted, difference, and filtered difference images along with pixel intensity distributions have been shown in Fig. S18. As shown in Fig. 6 (a), bulk images and those containing substitutional atoms exhibit significant visual similarity, differing only in subtle pixel intensity variations at the lattice sites of the substitutional atoms. This scenario presents a critical test for model performance. The same model previously employed for CdTe clustering was applied to $SrTiO_3$, and the initial analysis using all 47 computed features revealed the presence of bad features and the predicted N_c was three, deviating from the four intrinsic image categories, by putting bulk images and those with substitutional atoms in the same cluster as detailed in optimization plot in Fig. 6(b). The clustering accuracy at this stage was found to be 85.35%, with additional results provided in Supplementary Section H.

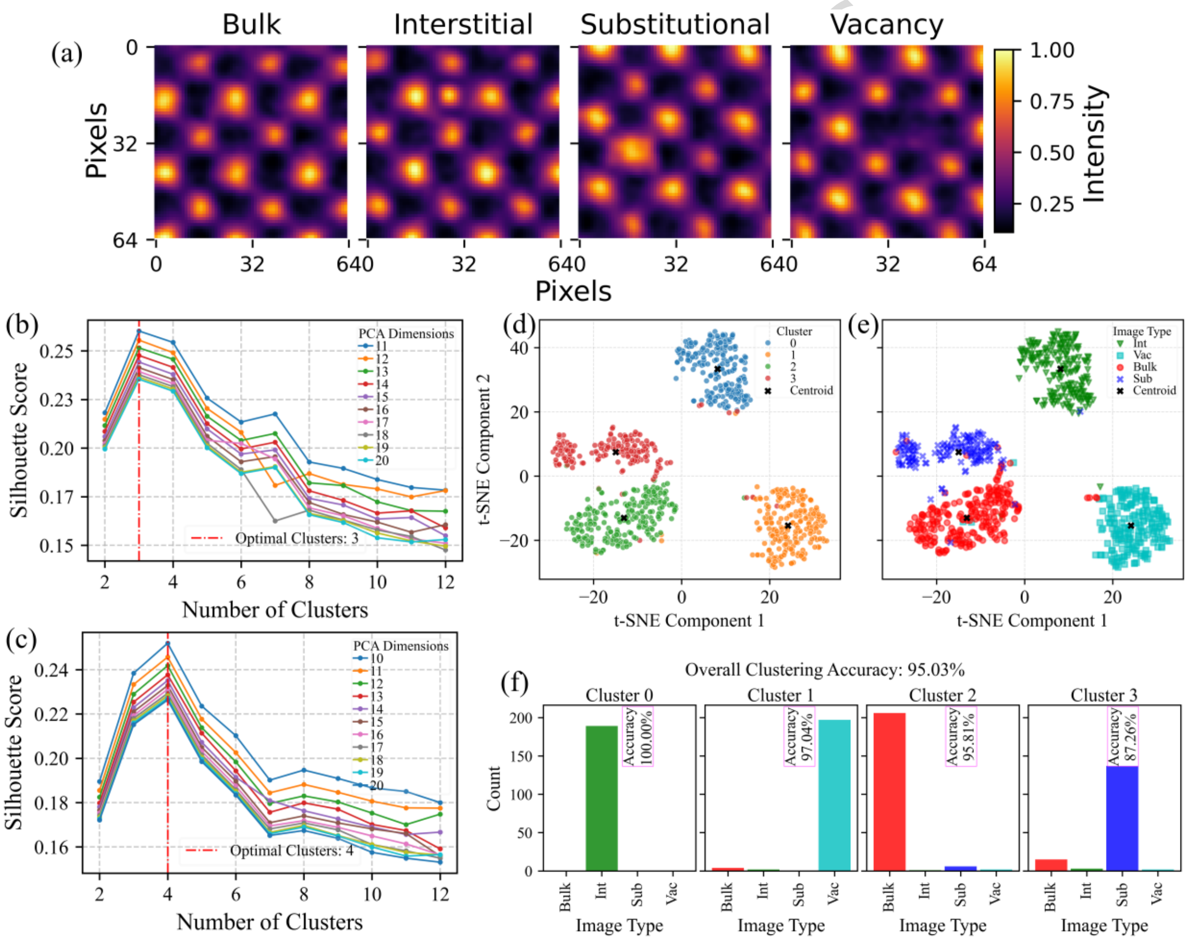

Figure 6: Clustering Analysis of the $SrTiO_3$ Dataset: Results are presented for 764 images classified into four distinct categories: bulk, interstitial atoms (Int), substitutional atoms (Sub), and vacancies (Vac). **(a)** Representative images from each category. **(b), (c)** The optimized PCA dimensions and number of clusters before and after applying three-tier feature selection process. **(d)** t-SNE visualization with data points colored according to cluster assignments, with black "x" symbols marking cluster centroids, demonstrating well-separated groupings. (**e)** t-SNE visualization based on image type, showing consistent centroid positions and confirming that similar images are grouped together. **(f)** Histogram of clustering accuracy, indicating an overall accuracy of 95.03%.

However, at this stage, the significance of the three-tier feature selection process becomes evident once again. After passing all 47 features from three feature shortlisting filters, the final set of 29 refined features, selected by the model, was subsequently utilized for optimizing p and k. Upon implementing the three-tier feature selection strategy developed in this model, N_c was correctly identified as four, as shown in Fig. 6(c). This result underscores the efficacy of our three-step feature selection approach in effectively eliminating redundant, noisy, and non-informative features. These optimized parameters were then employed in the final clustering process. The t-SNE visualizations of the $SrTiO_3$ dataset, presented in Figs. 6(d) and 6(e), illustrate the final clustering outcomes: Fig. 6(d) depicts cluster assignments using four distinct colors, each representing a unique cluster, while Fig. 6(e) employs color and marker variations based on image type, confirming that images of the same type were accurately grouped. The final clustering accuracy, depicted through histograms in Fig. 6(f), reached an improved value of 95.03% which was 85.35% before feature shortlisting. This validation of our clustering framework on the $SrTiO_3$ dataset highlights the consistency and flexibility of the unsupervised machine learning approach proposed in this study.

Performance Evaluation and Stability Analysis

To quantitatively evaluate the stepwise contribution of each module in the workflow, the clustering accuracies obtained after eliminating each critical step are summarized in Table 1, highlighting the importance of the CVAE module and the three-tier feature selection process.

Table 1: Quantitative evaluation of individual components in the proposed CVAE-based clustering framework. The table summarizes the overall clustering accuracy (%) for CdTe and $SrTiO_3$ datasets under different workflow configurations. Removing either the CVAE module or the three-tier feature selection step results in a marked decrease in clustering performance, confirming that each component contributes uniquely to the overall accuracy of the framework. Detailed clustering visualizations for the first two cases reported in the table are provided in Supplementary Sections E, G, and H.

Raw input images	CVAE (difference and filtered-difference images)	Three-tier feature selection	Overall clustering accuracy (%)	
			CdTe dataset	$SrTiO_3$ dataset
✓	✗	✗	73.80	30.68
✓	✓	✗	88.24	85.35
✓	✓	✓	98.46	95.03

To further validate the reliability of the clustering outcomes, a comprehensive stability analysis was conducted using the optimal number of PCA dimensions identified for each dataset, as shown in Fig. 7. Three complementary tests were performed to examine robustness against initialization randomness, centroid restarts, and data resampling. In Fig. 7, the top and bottom rows correspond to the stability analysis results for the CdTe and $SrTiO_3$ datasets, respectively. Figs. 7(a) and 7(d) demonstrate that the mean silhouette scores ± standard deviation ($\bar{S} \pm \sigma$) exhibit negligible variability across 100 random initializations ($n_{\mathrm{init}} = 1$). Figs. 7(b) and 7(e) show S as a function

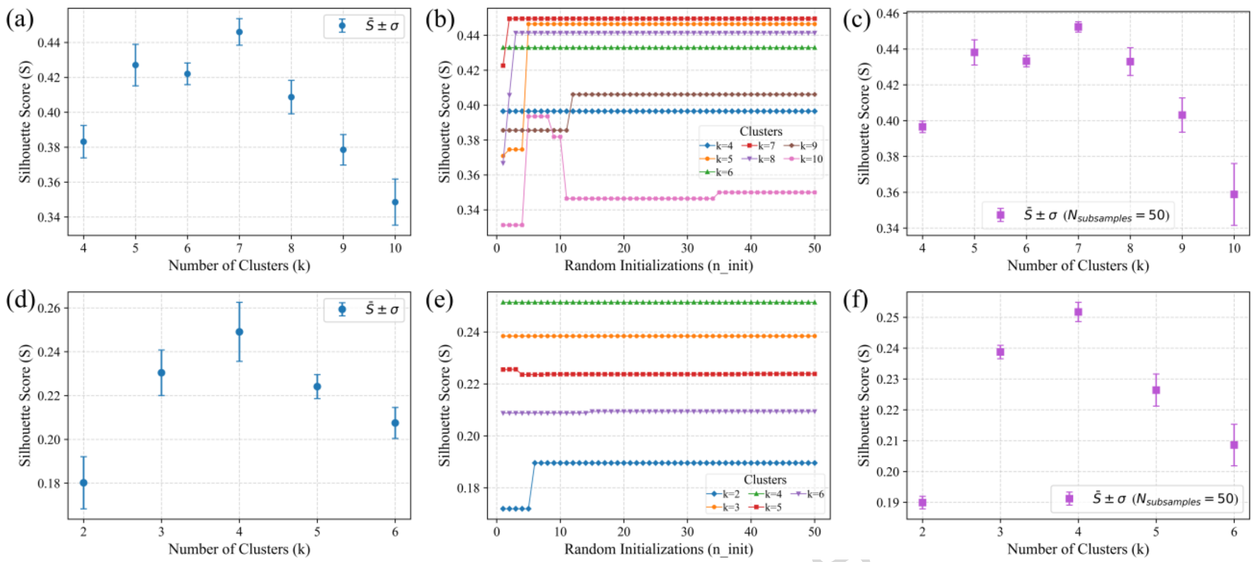

Figure 7: Initialization, multi-initialization, and subsample stability analyses for CdTe and $SrTiO_3$ datasets. (a–c) CdTe and **(d–f)** $SrTiO_3$. All analyses were performed using the optimal number of PCA dimensions determined for each dataset. **(a, d)** Initialization robustness across 100 random seeds ($n_{\text{init}} = 1$), showing mean ± standard-deviation error bars of the silhouette score ($\bar{S} \pm \sigma$) for each k. The narrow spreads indicate negligible sensitivity to random initialization. **(b, e)** Multi-initialization sweep ($n_{\text{init}} = 1$–50) demonstrating convergence of silhouette scores across repeated centroid restarts; the optimal $k = 7$ for CdTe and $k = 4$ for $SrTiO_3$ remain consistently highest. **(c, f)** Subsample stability analysis over 50 random 80 % subsets ($n_{\text{init}} = 10$), confirming that clustering performance is stable under data perturbations.

of 50 multi-initialization restarts ($n_{\text{init}} = 1$–50), while Figs. 7(c) and 7(f) present the $\bar{S} \pm \sigma$ values obtained from 50 random subsampling trials (80 % subsets, $n_{\text{init}} = 10$). The standard deviations remained below 0.015 in all cases, confirming that the identified cluster structures are statistically stable. In addition, pairwise win-rate analyses comparing the N_c with competing k values, provided in Supplementary Fig. S22, show that N_c achieved a dominance rate exceeding 94 % across random seeds. Collectively, these results establish that the proposed CVAE-based clustering framework yields not only superior quantitative accuracy but also high reproducibility and robustness with respect to initialization and data perturbation.

Discussion

In this study, a fully unsupervised machine learning framework for automated defect identification and clustering in atomic-resolution (S)TEM images is presented, eliminating the need for pre-labeled training datasets. By integrating automated feature selection, dimensionality reduction, and clustering optimization, the model achieves highly accurate defect grouping, even in cases where subtle structural variations or edge effects hinder conventional computer vision methods. The framework's effectiveness is validated through detailed cluster analysis, including t-SNE visualizations, mathematically formulated histograms, and stability tests under different random initializations and data perturbations, providing a comprehensive assessment of its clustering capability. The results highlight not only the precision of unsupervised defect clustering but also the robustness of the feature selection strategy in capturing meaningful structural variations. Regarding limitations of the current study, we note that the presence of multiple defects within a single image patch can introduce partial feature overlap, which may reduce cluster separability, particularly for larger patch sizes. Conversely, excessively small patches may limit structural context and reconstruction fidelity. As discussed in the Supplementary Sections A-iii and J, these effects can be mitigated through an appropriate choice of patch size that balances structural context and defect isolation.

Beyond unsupervised defect identification, this approach holds significant potential for AI-driven microscopy, enabling autonomous detection, clustering, and real-time analysis of structural anomalies. By efficiently analyzing large datasets, this unsupervised clustering framework can facilitate the statistical quantification of defect concentrations within crystalline structures, which in turn enables the autonomous exploration of defect-property relationships, accelerating materials characterization. Integration with high-throughput experimental workflows or autonomous synthesis platforms could further enhance self-optimizing materials design, advancing the development of self-driven laboratories. Future extensions may incorporate multi-modal datasets by integrating complementary spectroscopic and diffraction information. By leveraging unsupervised learning, this work contributes to the broader vision of intelligent, data-driven microscopy, ultimately paving the way for the next generation of automated, high-precision materials characterization.

Building on these future extensions, the proposed workflow opens clear opportunities for extending unsupervised CVAE-based clustering toward physically interpretable defect classification. In future studies, unsupervised cluster assignments can be used as pseudo-labels to train supervised models (e.g., Random Forest classifiers), enabling post hoc analysis of feature importance to identify which descriptors most strongly distinguish specific clusters. In parallel, human-in-the-loop inspection of representative input, reconstructed, and filtered-difference images from each cluster can support the assignment and validation of physical defect types. Together, these extensions provide a systematic pathway for connecting unsupervised clustering outcomes to interpretable defect signatures, advancing explainable and physically grounded analysis of atomic-resolution microscopy data.

Methods

CVAE Architecture and Training Setup

The CVAE implemented in this study comprises two primary components: an encoder and a decoder, designed as architectural mirrors of each other. Each branch consists of three convolutional layers having 64, 128, and 128 filters, respectively (Fig. 1b). The model was trained on a single high-resolution STEM image in TIFF format. To construct the training dataset, 100 image patches of size 64 × 64 pixels were randomly extracted from the lower half of the image. The patch size was selected to be slightly larger than a single unit cell, ensuring the inclusion of relevant local structural features (see Supplementary Section A-iii for the patch-size sensitivity analysis of the model). Prior to training, a Gaussian blur with $\sigma = 1.0$ was applied to each patch to suppress shot and scan noise, followed by normalization of pixel intensities to the range [0, 1]. The CVAE model was trained over 300 epochs using a batch size of 16. Model performance was evaluated using the evidence lower bound (ELBO), computed on a separate test set of 100 patches randomly extracted from the upper half of the same STEM image. This metric reflects both reconstruction fidelity and the quality of the learned latent representation. Additionally, reconstructed outputs were visually compared to the original inputs to qualitatively assess the model's ability to capture atomic-scale structural features. For further details on the training procedure and theoretical underpinnings of the CVAE model, we refer the reader to our earlier publication by Enea et al.[31], where the methodological framework is discussed in depth. In addition, complete details for training the model, along with the training images, and executing the clustering assignments specific to this study are provided on the author's GitHub repository at https://github.com/RAW-meh/defect-classification-in-stem-images.

Test Dataset Generation for Clustering

After training, the CVAE model can be directly used to analyze 64 × 64 pixel images, generating corresponding latent representations, reconstructed outputs, and filtered difference maps. To construct the test dataset for clustering, image patches were extracted from various regions of the STEM image (Fig. 2), each corresponding to a specific defect type. For instance, to simulate vacancy-type defects, artificial dumbbell vacancies were introduced at different positions within the image shown in Fig. 2(a). From these modified regions, patches (64 × 64 pixels) were extracted using a stride of 5-20 pixels, capturing a range of defect configurations and edge-related effects. This procedure was repeated to generate representative data for seven distinct image classes. The same model training and dataset construction methodology was also applied to the $SrTiO_3$ dataset to ensure consistency and cross-material applicability.

Implementation Details

All data processing and model implementation were conducted using custom Python scripts (version 3.12.3), executed within Jupyter Notebook (version 6.5.7). The CVAE model was developed using TensorFlow (version 2.18.0)[45]. Clustering and dimensionality reduction tasks, including k-means and PCA, were performed using the scikit-learn library (version 1.5.1)[46]. All source codes, training images, test datasets, and detailed usage instructions are publicly available in the authors' GitHub repository at https://github.com/RAW-Ayyubi/defect-classification-in-stem-images.

Data Availability

The data and Python codes supporting the findings of this study are openly available on GitHub at https://github.com/RAW-Ayyubi/defect-classification-in-stem-images.

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Author contributions

R.F.K. and J.P.B. conceived the project and supervised the research. R.A.W.A. conducted the investigation, developed the software, generated the visualizations, and prepared the original draft of the manuscript. S.S. contributed to formal analysis and validation. J.P.B. and R.A.W.A. carried out methodology development. R.F.K. and J.P.B. contributed to conceptualization. R.F.K. led funding acquisition, project administration, and resource provision. R.F.K. further shared supervision responsibilities. All authors reviewed and commented on the final version of the manuscript.

Competing interests

The authors declare no competing interests.

Additional Information
Supporting Information is available.

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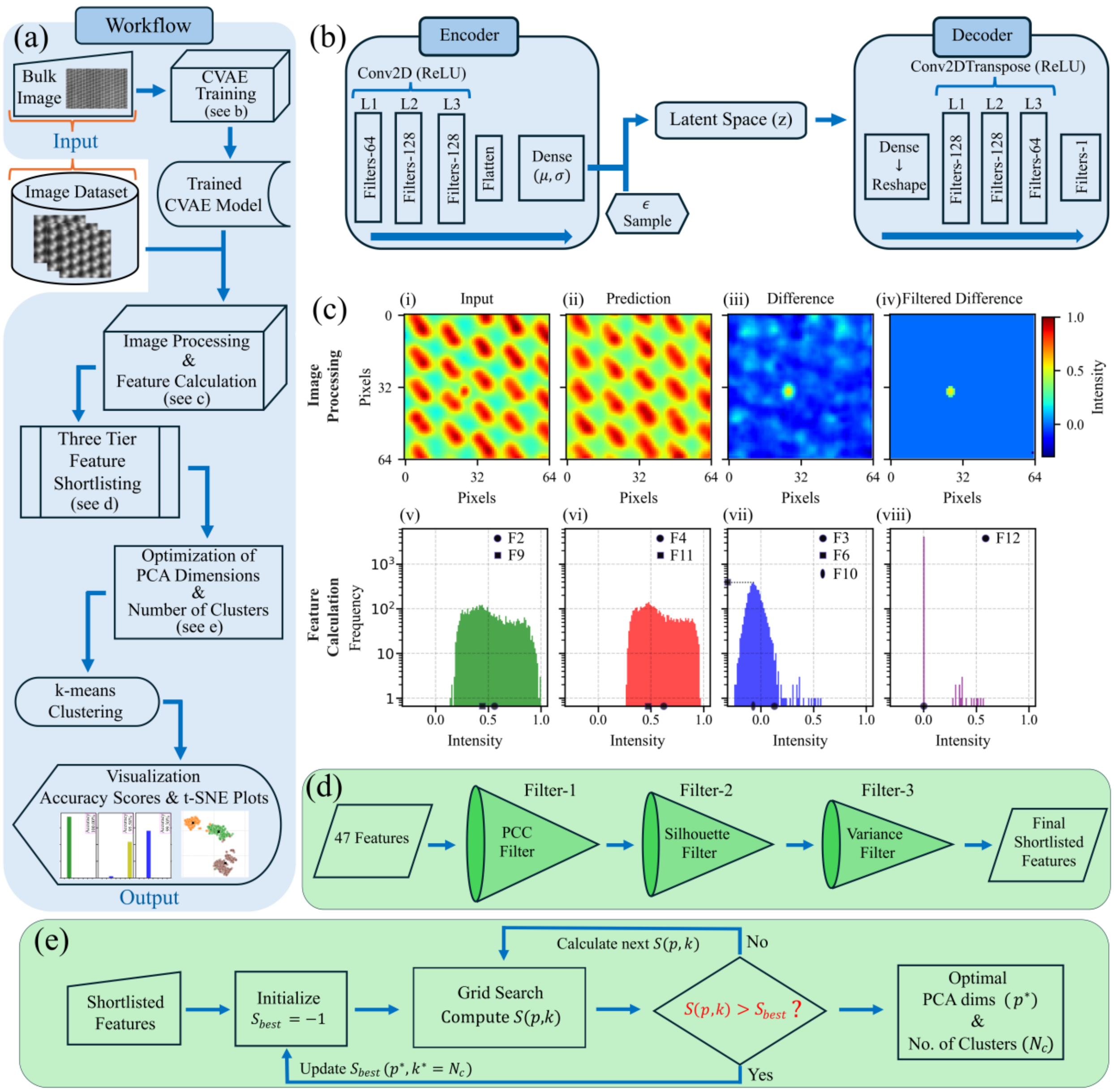

Figure 1: Overview of the clustering pipeline. (a) Flowchart illustrating the complete pipeline, from CVAE training on periodic lattice structures to clustering based on identified anomalies. **(b)** CVAE architecture comprising an encoder-decoder pair used to learn latent representations of the periodic structures. **(c)** Illustration of how the trained CVAE iteratively processes all images in the dataset, exemplified by an image containing an interstitial atom: (i) Colored grayscale input, (ii) defect-free reconstruction by the CVAE, (iii) difference image, and (iv) noise-filtered difference obtained by thresholding. (v–viii) Corresponding histograms of pixel-intensity distributions for the input, predicted, difference, and filtered difference images, used to extract 47 quantitative features. **(d)** Three-tier feature shortlisting process, removing noisy and uninformative features before clustering. **(e)** Workflow for determining the optimal number of clusters (N_c) and PCA dimensions (p^*) by maximizing the silhouette score (S).

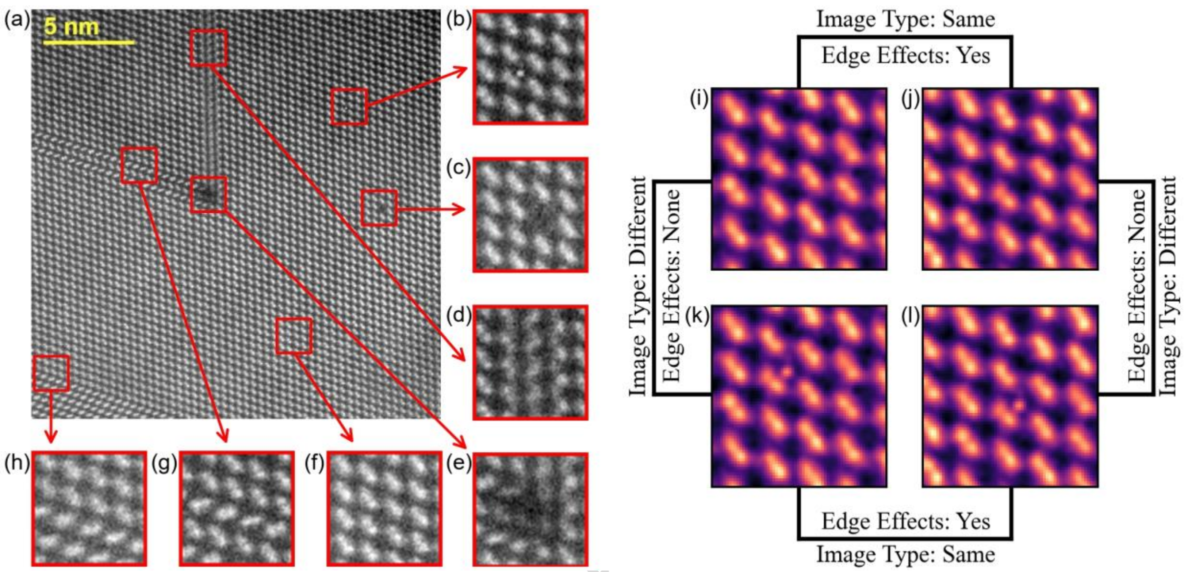

Figure 2: Representative Atomic-Resolution STEM Image of CdTe with Annotated Defects and Edge Effects. (a) Atomic-resolution high-angle annular dark field (HAADF) image of CdTe along the [110] crystallographic direction. The scale bar represents 5 nm. The HAADF STEM image was acquired using an aberration-corrected cold field emission scanning transmission electron microscope operated at an accelerating voltage of 200 kV. The field of view spans approximately 22 × 22 nm, where the CdTe dumbbells are distinctly discernible, exhibiting a pronounced bright contrast. The inner and outer collection angles for the HAADF detector were set to 68 mrad and 280 mrad, respectively. The image has dimensions of 768 × 768 pixels, with seven distinct regions highlighted by red boxes of size 64 × 64 pixels. The zoomed-in views of these regions are presented in **(b)** an interstitial atom, **(c)** a missing CdTe dumbbell, **(d)** a type-1 stacking fault, **(e)** a Lomer-Cottrell dislocation, **(f)** a defect-free bulk region, **(g)** a type-2 stacking fault, and **(h)** a twin boundary. **(i)–(l)** illustrate the presence of translation-induced edge effects in the images used for clustering analysis: **(i)** and **(j)** correspond to the same image type (defect-free bulk region) but with different edge effects, while **(k)** and **(l)** represent the same image type (interstitial atom) but with varying edge effects.

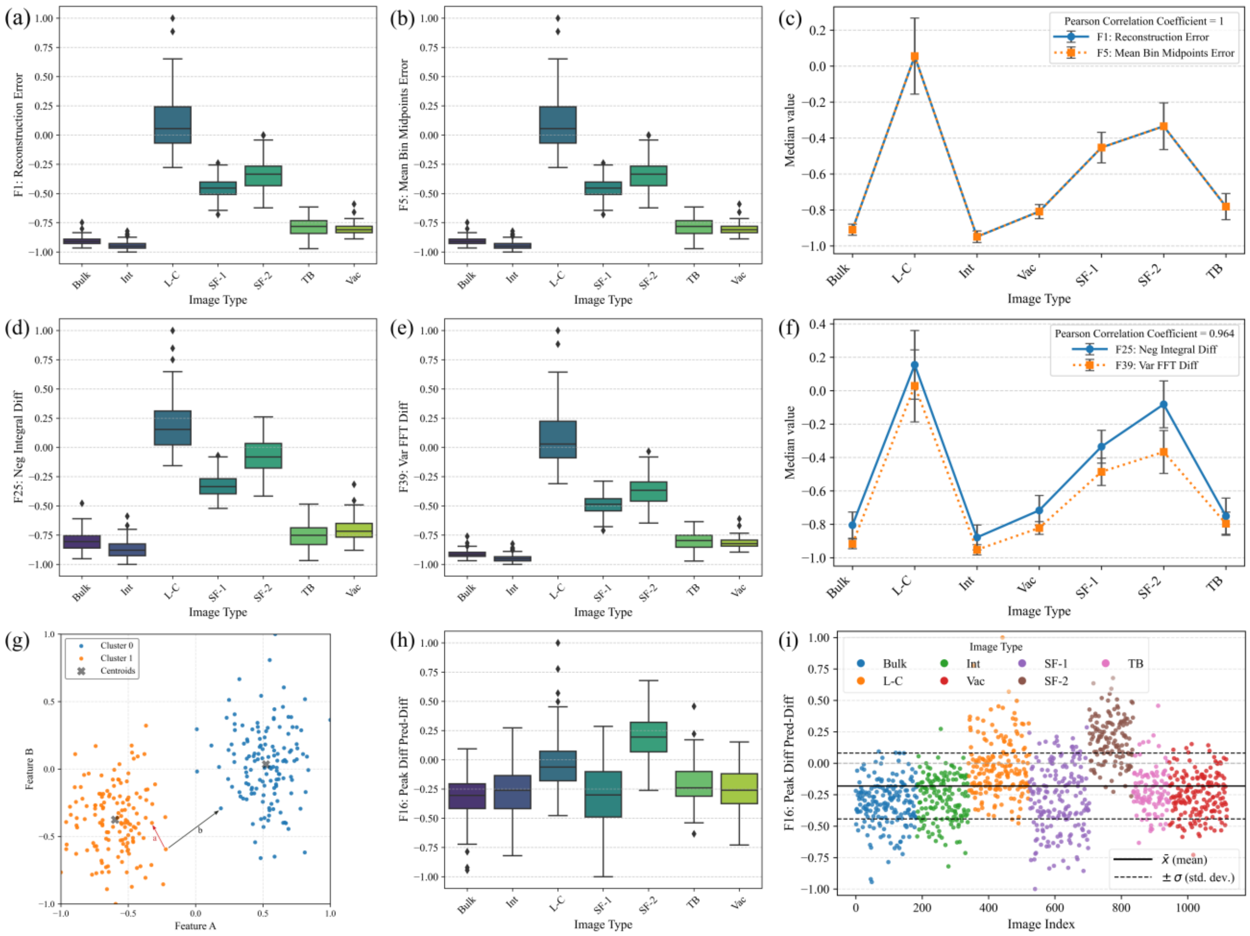

Figure 3: Feature filtering analysis. (a,b) Box-and-whisker plots of features F1 and F5 showing identical distributions, indicating redundant information content. **(c)** Linear-dependency plot for F1 and F5 with correlation coefficient $r = 1$. **(d,e)** Box-and-whisker plots of features F25 and F39 displaying similar trends, confirmed by their near-perfect linear correlation in **(f)** ($r = 0.96$). **(g)** Illustration of the silhouette-score geometry, where a represents the average intra-cluster distance and b the smallest average inter-cluster distance used in Eq. (3). **(h,i)** Box-and-whisker and value-distribution plots of feature F16, illustrating a narrow variance range ($\pm\,\sigma = 0.32$) across all image types, indicating negligible contribution to clustering.

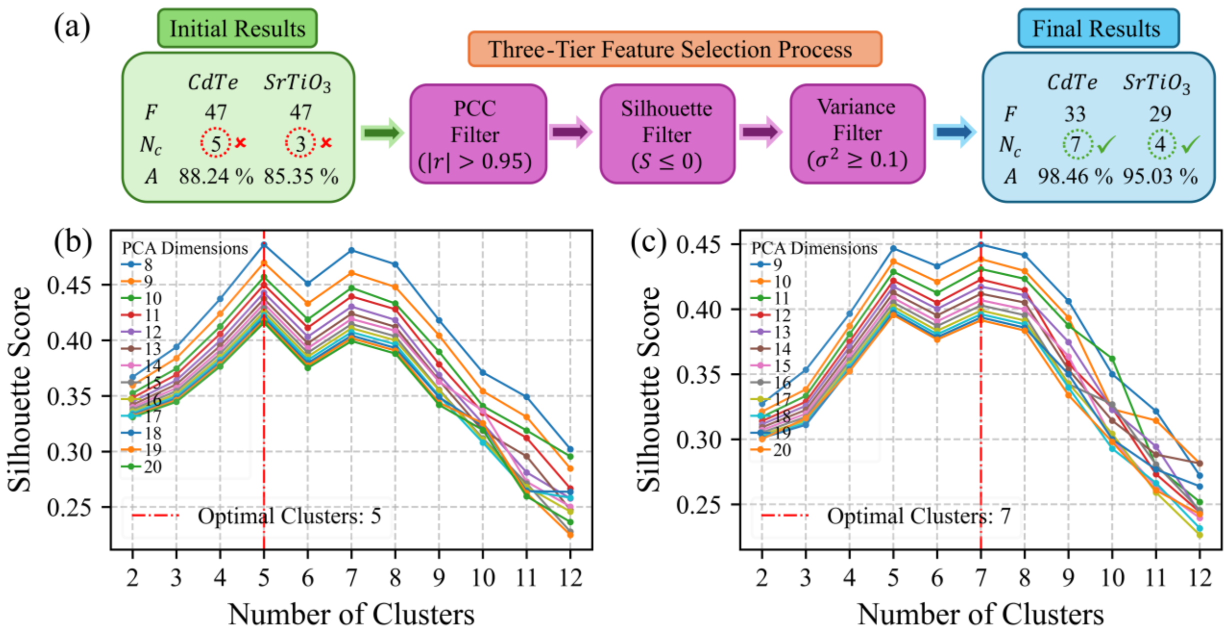

Figure 4: Feature Selection and Automated Clustering Optimization for Atomic-Resolution STEM Images. (a) Schematic of the feature selection pipeline, illustrating descriptor refinement and clustering enhancement for atomic-resolution STEM images. The flowchart tables summarize results before and after three-tier feature selection, where F is the number of features, N_c is the optimal number of clusters, and A represents overall clustering accuracy. **(b), (c)** Automated optimization process for determining the optimal PCA dimensions and number of clusters for the CdTe dataset, before and after the three-tier feature selection process, respectively, utilizing silhouette score maximization as the key criterion. The red vertical dashed lines indicate the optimal number of clusters.

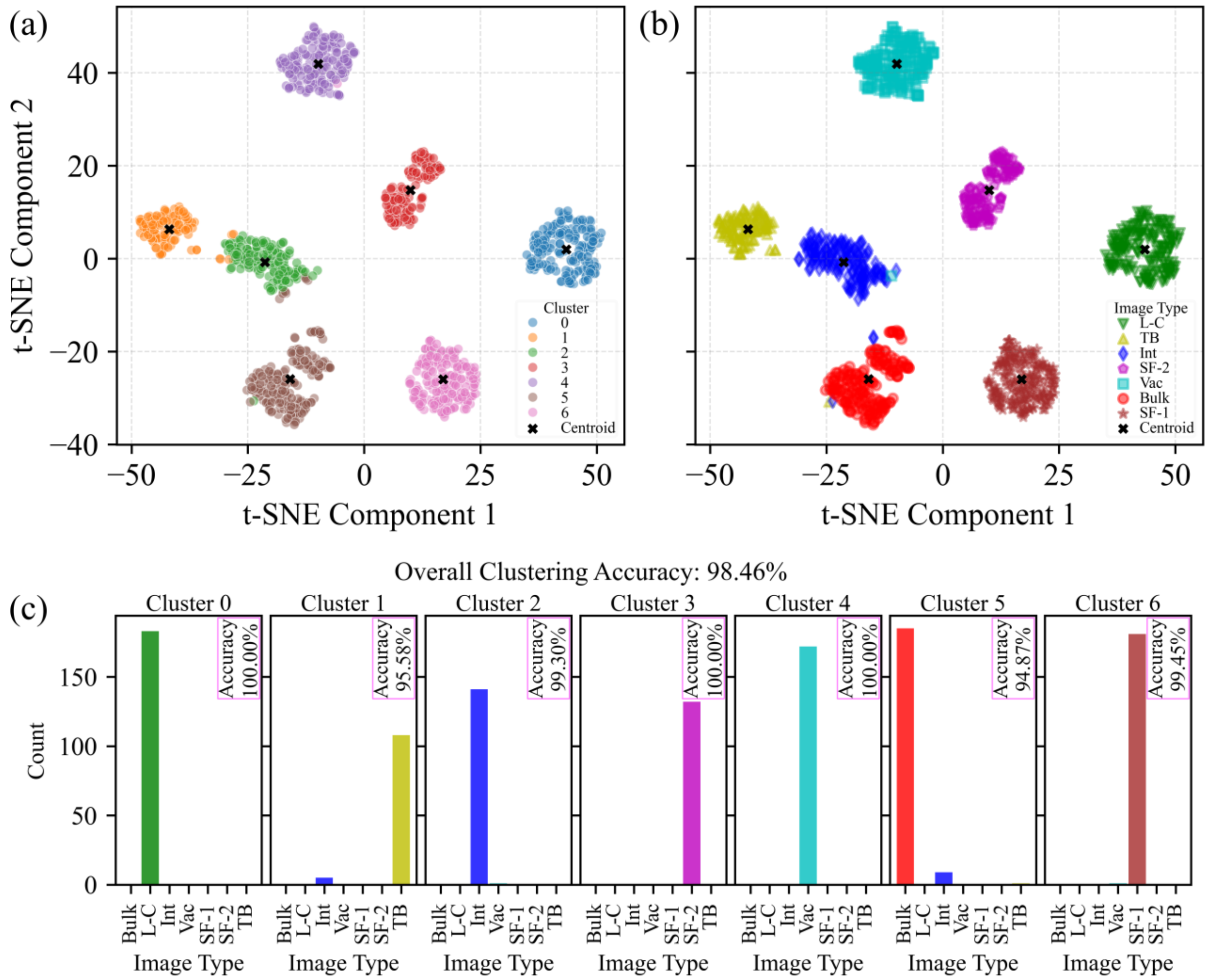

Figure 5: t-SNE visualization of feature distributions for a dataset comprising 1120 atomic-resolution STEM images of CdTe, categorized into seven distinct image types. (a) Clustering-based t-SNE representation, where data points are colored according to their assigned clusters. Cluster centroids are marked with black "x" symbols, illustrating well-separated feature groupings. **(b)** Image-type-based t-SNE representation, where data points are assigned colors and markers corresponding to their image types, confirming that similar images are grouped within the same cluster. Centroids remain consistent across both visualizations, underscoring structural similarities. **(c)** Histogram representation of clustering accuracy, demonstrating an overall accuracy of 98.46%.

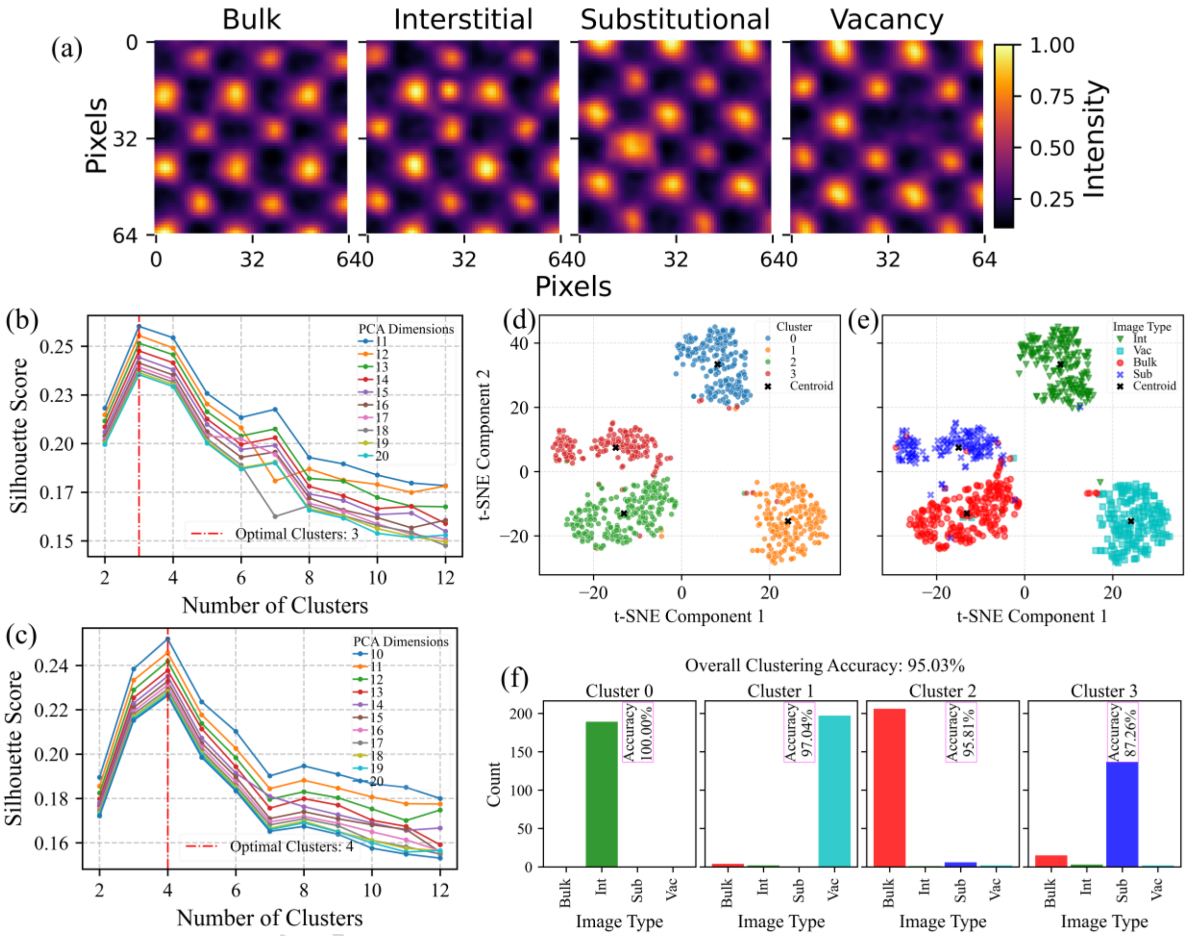

Figure 6: Clustering Analysis of the $SrTiO_3$ Dataset: Results are presented for 764 images classified into four distinct categories: bulk, interstitial atoms (Int), substitutional atoms (Sub), and vacancies (Vac). **(a)** Representative images from each category. **(b), (c)** The optimized PCA dimensions and number of clusters before and after applying three-tier feature selection process. **(d)** t-SNE visualization with data points colored according to cluster assignments, with black "x" symbols marking cluster centroids, demonstrating well-separated groupings. (**e**) t-SNE visualization based on image type, showing consistent centroid positions and confirming that similar images are grouped together. **(f)** Histogram of clustering accuracy, indicating an overall accuracy of 95.03%.

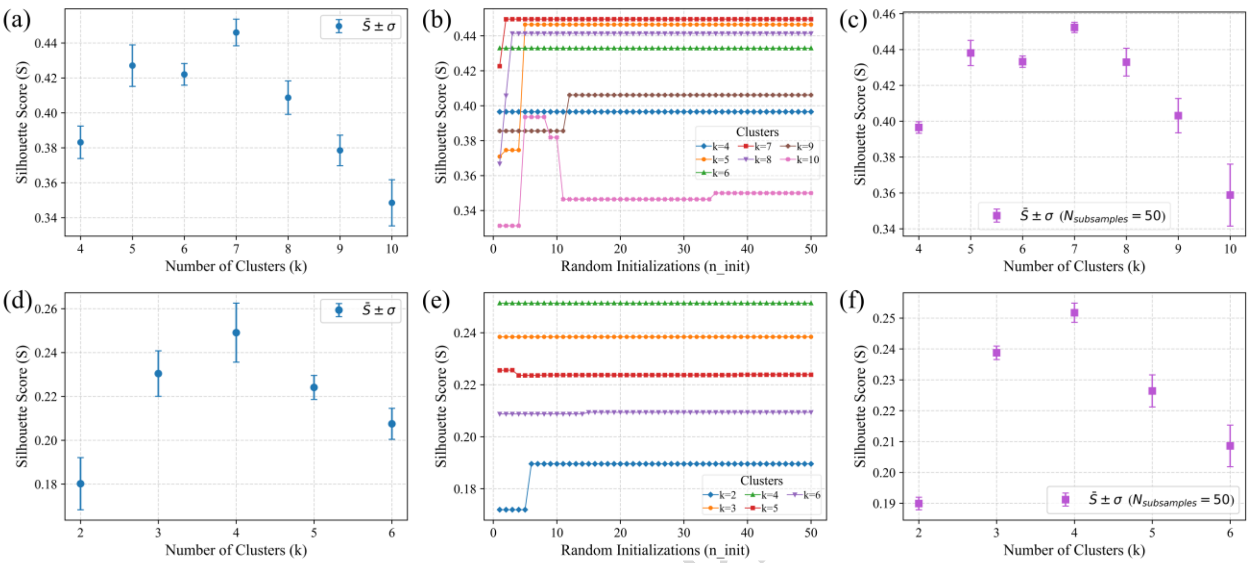

Figure 7: Initialization, multi-initialization, and subsample stability analyses for CdTe and $SrTiO_3$ datasets. (a–c) CdTe and **(d–f)** $SrTiO_3$. All analyses were performed using the optimal number of PCA dimensions determined for each dataset. **(a, d)** Initialization robustness across 100 random seeds ($n_{\text{init}} = 1$), showing mean ± standard-deviation error bars of the silhouette score ($\bar{S} \pm \sigma$) for each k. The narrow spreads indicate negligible sensitivity to random initialization. **(b, e)** Multi-initialization sweep ($n_{\text{init}} = 1$–50) demonstrating convergence of silhouette scores across repeated centroid restarts; the optimal $k = 7$ for CdTe and $k = 4$ for $SrTiO_3$ remain consistently highest. **(c, f)** Subsample stability analysis over 50 random 80 % subsets ($n_{\text{init}} = 10$), confirming that clustering performance is stable under data perturbations.

Raw input images	CVAE (difference and filtered-difference images)	Three-tier feature selection	Overall clustering accuracy (%)	
			CdTe dataset	$SrTiO_3$ dataset
✓	✗	✗	73.80	30.68
✓	✓	✗	88.24	85.35
✓	✓	✓	98.46	95.03

Table 1: Quantitative evaluation of individual components in the proposed CVAE-based clustering framework. The table summarizes the overall clustering accuracy (%) for CdTe and $SrTiO_3$ datasets under different workflow configurations. Removing either the CVAE module or the three-tier feature selection step results in a marked decrease in clustering performance, confirming that each component contributes uniquely to the overall accuracy of the framework. Detailed clustering visualizations for the first two cases reported in the table are provided in Supplementary Sections E, G, and H.